

EE1060: Differential Equations and Transform Techniques**Solutions to Practice Problem Set - 05**

1. Proof is similar to the one covered in the class for even functions
2. Begin by using the fact that an even function can be expressed as $g(x) = \frac{1}{2} [g(x) + g(-x)]$. Then, represent both $g(x)$ and $g(-x)$ using the Fourier synthesis equation. Rearrange the resulting terms to demonstrate that the imaginary part of C_k is zero i.e., $\text{Imag}\{C_k\} = 0$.
3. Begin by using the fact that an odd function can be expressed as $g(x) = \frac{1}{2} [g(x) - g(-x)]$. Then, represent both $g(x)$ and $g(-x)$ using the Fourier synthesis equation. Rearrange the resulting terms to demonstrate that the real part of C_k is zero i.e., $\text{Real}\{C_k\} = 0$.
4. Begin with the expressions for Fourier Coefficients when the function is half-wave symmetric. Use the additional condition $g\left(\frac{L}{4} - \theta\right) = g\left(\frac{L}{4} + \theta\right)$ to derive the required result.
5.
$$f_1(x) = \frac{A}{2} + \sum_{k=1, k=\text{odd}}^{\infty} \frac{4A}{\pi k} \sin(k\omega_0 x)$$
6.
$$f_2(x) = \frac{A}{2} - \sum_{k=1, k=\text{odd}}^{\infty} \frac{4A}{\pi^2 k^2} \cos(k\omega_0 x)$$
7.
$$C_n = \begin{cases} \frac{A}{\pi n} \sin\left(\frac{2\pi n \alpha}{L}\right), & n \neq 0, \\ \frac{2A\alpha}{L}, & n = 0. \end{cases}$$
8.
$$f_4(x) = \frac{2A}{\pi} + \sum_{k=1, k=\text{odd}}^{\infty} \frac{2A}{\pi k} \cos(k\omega_0 x)$$
9. Question on Legendre Polynomials:
 - (a) It can easily be shown that the inner product between any two functions in the set $\left\{p_0(x), p_1(x), p_2(x) = \frac{1}{2}(3x^2 - 1)\right\}$ is 0 in the interval $[-1, 1]$.
 - (b) $f(x) = \frac{2}{5}x$
 - (c) $h(x) = \frac{2}{5}x + \frac{2}{35}(5x^3 - 3x)$.
10. The given set of functions are orthogonal on the given interval. The projection is $\frac{2}{5}$.